

Midterm 1 **Solutions**

EECS 245, Fall 2026 at the University of Michigan

Name: _____

username: _____

UMID: _____

Room: ☐ 1013 DOW ☐ 1014 DOW (alternate/SSD) ☐ 3901 BBB (alternate)

Instructions

- This exam consists of 8 problems, worth a total of 100 points, spread across 19 pages.
- You have 120 minutes to complete this exam, unless you have extended-time accommodations through SSD.
- Write your username in the top right corner of each page where a space is provided.
- When a large box is provided, **you must show all of your work and justify your thinking** there. Write your final answer in the small box, if one is provided. We will not grade work that appears outside the space provided. When justification is required, answers without sufficient justification will not receive full credit.
- For multiple choice problems, completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.
 - ☐ A bubble means that you should only select one choice.
 - ☐ A square box means that you should select all that apply.
- You may refer to **one double-sided 8.5x11" handwritten notes sheet**. Other than that, you may not refer to any other resources or technology during the exam (no phones, watches, or calculators).

You are to abide by the University of Michigan/Engineering Honor Code. To receive a grade, please sign below to signify that you have kept the Honor Code pledge.

I have neither given nor received aid on this exam, nor have I concealed any violations of the Honor Code.

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Tip: Skim through the entire exam before starting to work on it.

Problem 1 (12 pts)

Consider two datasets, $A = \{y_1, y_2, \dots, y_n\}$ and $B = \{z_1, z_2, \dots, z_m\}$. Let $R_A(w)$ and $R_B(w)$ represent the mean squared errors of the constant prediction w on datasets A and B , respectively:

$$R_A(w) = \frac{1}{n} \sum_{i=1}^n (y_i - w)^2 = (7 - w)^2 + 5, \quad R_B(w) = (13 - w)^2 + 17$$

a) (3 pts) Which dataset below could be A ?

- ☐ 7, 8, 10
 ☒ 4, 6, 8, 10
 ☐ 5, 6, 7, 8, 9
 ☐ 2, 3, 6, 7, 8

Solution: Recall that the mean squared error of a constant prediction can be written as

$$R_A(w) = (w - \bar{y})^2 + \sigma_y^2$$

So, the dataset must have a mean of 7 and a variance of 5. The dataset 4, 6, 8, 10 satisfies both requirements:

$$\bar{y} = \frac{4 + 6 + 8 + 10}{4} = 7, \quad \sigma_y^2 = \frac{9 + 1 + 1 + 9}{4} = 5$$

The dataset 5, 6, 7, 8, 9 also has a mean of 7, but its variance is 2, so it does not work.

b) (3 pts) Let $S(w)$ be the **sum** of the two mean squared error functions:

$$S(w) = R_A(w) + R_B(w)$$

Find w^* , the value of w that minimizes $S(w)$. Give your answer as a number with no variables.

Solution: Add the two mean squared errors and take the derivative:

$$S(w) = (7 - w)^2 + 5 + (13 - w)^2 + 17$$

$$\frac{d}{dw} S(w) = 2(w - 7) + 2(w - 13) = 4w - 40$$

Setting this equal to zero gives $w^* = 10$. The derivative is negative below 10 and positive above 10, so this is the minimizer.

This is the same kind of optimization we used to minimize mean squared error: the optimal prediction balances the two squared deviations. The constants +5 and +17 disappear when we take the derivative, so they do not affect the minimizer.

c) (6 pts) Let $M(w)$ return the **larger** of the two mean squared error functions:

$$M(w) = \max(R_A(w), R_B(w))$$

Find w^* , the value of w that minimizes $M(w)$. Give your answer as a number with no variables. *Hint: Draw a picture — a clearly annotated picture, accompanied by some algebra, is sufficient justification.*

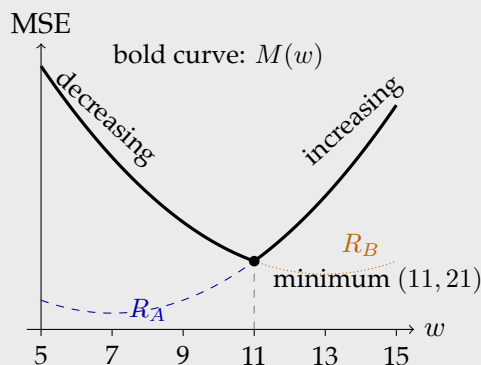
Solution: First, find where the two curves intersect:

$$\begin{aligned}(w - 7)^2 + 5 &= (w - 13)^2 + 17 \\ w^2 - 14w + 54 &= w^2 - 26w + 186 \\ 12w &= 132 \\ w &= 11\end{aligned}$$

Since $R_A(w) - R_B(w) = 12(w - 11)$, when $w > 11$, we have $R_A(w) > R_B(w)$, so $M(w)$ returns $R_A(w)$. Otherwise, $M(w)$ returns $R_B(w)$; at $w = 11$, the two are equal. Thus,

$$M(w) = \begin{cases} R_B(w), & w \leq 11 \\ R_A(w), & w \geq 11 \end{cases}$$

R_B is decreasing for $w < 11$, since its vertex is at 13. R_A is increasing for $w > 11$, since its vertex is at 7. Therefore, M decreases up to 11 and increases after 11, giving $w^* = 11$ and $M(11) = 21$.



Problem 2 (14 pts)

Consider a dataset of seven values $y_1, y_2, \dots, y_7$, listed in **non-decreasing** order:

$$y_1, y_2, 47, 60, 60, 81, y_7$$

Suppose that the mean of $y_1, \dots, y_6$ — that is, **not including** y_7 — is 50. Furthermore, suppose

- $f(w) = \frac{1}{7} \sum_{i=1}^7 |y_i - w|$, and α^* is the value of w that minimizes $f(w)$.
- $g(w) = \frac{1}{7} \sum_{i=1}^7 (y_i - w)^2$, and β^* is the value of w that minimizes $g(w)$.

a) (2 pts) What is the value of α^* ? Give your answer as a number with no variables.

$$\alpha^* = \boxed{60}$$

Solution: Mean absolute error is minimized by the median. Since there are seven ordered values, the median is the fourth value, so $\alpha^* = 60$. The fact that 60 is repeated does not change this result: the middle (fourth) value is still 60, so the median is still 60.

b) (5 pts) What is the largest possible value of y_7 such that $\alpha^* \geq \beta^*$? Give your answer as a number with no variables.

Solution: Mean squared error is minimized by the mean. The first six values sum to $6 \cdot 50 = 300$, so

$$\beta^* = \frac{300 + y_7}{7}$$

Using $\alpha^* = 60$, the required condition becomes

$$60 \geq \frac{300 + y_7}{7} \iff 420 \geq 300 + y_7 \iff y_7 \leq 120$$

The value 120 is consistent with the ordering, so the largest possible value is $\boxed{120}$.

c) (7 pts) Consider the same seven values, listed in **non-decreasing** order:

$$y_1, y_2, 47, 60, 60, 81, y_7$$

Recall,

$$f(w) = \frac{1}{7} \sum_{i=1}^7 |y_i - w|$$

For this part, do not assume that $\alpha^* \geq \beta^*$.

Suppose that $f(59) = 30$. What is $f(65)$? Give your answer as a number with no variables.

Hint: Consider the slopes of $f(w)$ between $w = 59$ and $w = 65$.

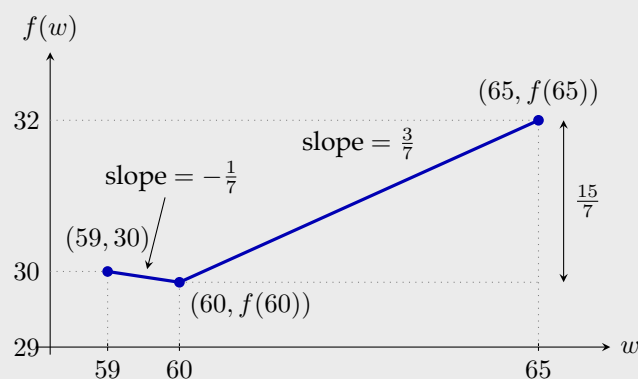
Solution: For any value of w that is not equal to a y_i , the slope of mean absolute error, $f(w)$, is

$$\frac{d}{dw} f(w) = \frac{\# \text{ of points left of } w - \# \text{ of points right of } w}{7}$$

For $59 < w < 60$, there are three values to the left and four to the right, so the slope is $-1/7$. For $60 < w < 65$, there are five to the left and two to the right, so the slope is $3/7$. Both copies of 60 count.

$$f(60) = 30 + (60 - 59) \left(-\frac{1}{7} \right) = \frac{209}{7}$$

$$f(65) = f(60) + (65 - 60) \left(\frac{3}{7} \right) = \frac{209}{7} + \frac{15}{7} = \boxed{32}$$



Problem 3 (16 pts)

Consider a dataset of n points, $(x_1, y_1), \dots, (x_n, y_n)$, where not all of the x_i are the same. We'd like to fit a **modified** simple linear regression model whose slope and intercept are forced to be the same:

$$h(x_i) = w + wx_i$$

The value of w that minimizes mean squared error for this model is

$$w^* = \frac{\sum_{i=1}^n (1 + x_i)y_i}{\sum_{i=1}^n (1 + x_i)^2}$$

a) (3 pts) Suppose:

- $h^*(x_i) = w^* + w^*x_i$ is the fitted **modified** model, where w^* minimizes mean squared error.
- $g^*(x_i) = w_0^* + w_1^*x_i$ is the **regular** simple linear regression model fitted to the same dataset, where w_0^* and w_1^* are chosen to minimize mean squared error.

Fill in the ??? with the relationship that is **guaranteed**:

$$\frac{1}{n} \sum_{i=1}^n (y_i - g^*(x_i))^2 \quad \boxed{???} \quad \frac{1}{n} \sum_{i=1}^n (y_i - h^*(x_i))^2$$

☐ $>$
 ☐ $\geq$
 ☐ $=$
 ☐ $<$
 ☒ $\leq$

Solution: The relationship is $\leq$.

Think of regular simple linear regression as being more flexible: it *can* have the same slope and intercept if that minimizes MSE, and it can have different slope and intercept if that minimizes MSE even further. Anything the modified model can do, the regular model can do, and more. So the regular model's minimum MSE cannot be larger.

- b) (6 pts) Let r be the correlation coefficient between the x - and y -values, and let σ_x and σ_y be their standard deviations, respectively. Suppose

$$r = -\frac{3}{5}, \quad \sigma_x = 4, \quad \sigma_y = 10$$

Give each answer as a number with no variables. If it is not possible to determine an answer from the information given, write **N/A** in the box.

- (i) (3 pts) What is w^* , the optimal slope for the **modified** simple linear regression model?

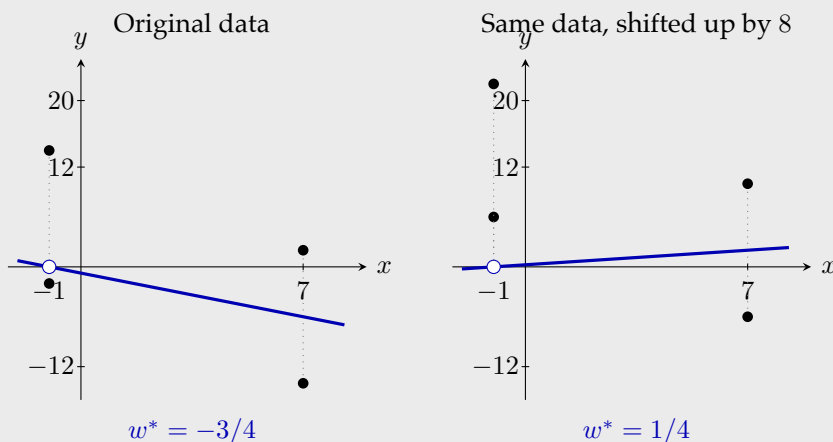
$$w^* = \boxed{\text{N/A}}$$

- (ii) (3 pts) What is w_1^* , the optimal slope for the **regular** simple linear regression model?

$$w_1^* = \boxed{-\frac{3}{2}}$$

Solution: (i) $\boxed{\text{N/A}}$. Moving the data changes which modified line fits best, even if its shape, correlation, and standard deviations stay the same. Every modified line $y = w + wx$ passes through $(-1, 0)$, so it cannot simply move along with the data.

For example, use x -values $-1, -1, 7, 7$ and y -values $-2, 14, -14, 2$ in the left panel. In the right panel, add 8 to every y -value. Both datasets have $r = -3/5$, $\sigma_x = 4$, and $\sigma_y = 10$.



At $x = -1$, the prediction is always 0, regardless of w . At $x = 7$, the prediction is $8w$, so the best choice makes $8w$ equal to the mean of the two y -values there. This gives $w^* = -3/4$ on the left and $w^* = 1/4$ on the right.

(ii) For regular simple linear regression,

$$w_1^* = r \frac{\sigma_y}{\sigma_x} = -\frac{3}{5} \cdot \frac{10}{4} = \boxed{-\frac{3}{2}}$$

- c) (7 pts) Show that w^* , the value of w that minimizes mean squared error for the modified simple linear regression model $h(x_i) = w + wx_i$, is

$$w^* = \frac{\sum_{i=1}^n (1 + x_i)y_i}{\sum_{i=1}^n (1 + x_i)^2}$$

Solution: The mean squared error of the modified model is

$$R(w) = \frac{1}{n} \sum_{i=1}^n (y_i - w(1 + x_i))^2$$

Taking the derivative with respect to w gives

$$R'(w) = -\frac{2}{n} \sum_{i=1}^n (1 + x_i)(y_i - w(1 + x_i))$$

Set the derivative equal to zero and solve for w :

$$\begin{aligned} \sum_{i=1}^n (1 + x_i)y_i - w \sum_{i=1}^n (1 + x_i)^2 &= 0 \\ w \sum_{i=1}^n (1 + x_i)^2 &= \sum_{i=1}^n (1 + x_i)y_i \end{aligned}$$

$$w^* = \frac{\sum_{i=1}^n (1 + x_i)y_i}{\sum_{i=1}^n (1 + x_i)^2}$$

The objective is a quadratic in w with a positive coefficient on w^2 , so this critical point is its minimum. **A second derivative test is not necessary.**

Problem 4 (11 pts)

Suppose a , b , and c are real numbers, and let

$$\vec{u} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} b \\ c \\ a \end{bmatrix}$$

- a) (5 pts) Use the Cauchy–Schwarz inequality to prove that the following holds for all real numbers a , b , and c :

$$|ab + bc + ca| \leq a^2 + b^2 + c^2$$

Solution: The Cauchy–Schwarz inequality says that

$$|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \|\vec{v}\|$$

For the given vectors,

$$\vec{u} \cdot \vec{v} = ab + bc + ca, \quad \|\vec{u}\| = \|\vec{v}\| = \sqrt{a^2 + b^2 + c^2}$$

By the Cauchy–Schwarz inequality,

$$|ab + bc + ca| = |\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \|\vec{v}\| = \boxed{a^2 + b^2 + c^2}$$

- b) (6 pts) Now suppose that $a + b + c = 0$ and $\vec{u} \neq \vec{0}$. Find θ , the angle between $\vec{u}$ and $\vec{v}$. Give your answer as the cosine inverse of a number (e.g. $\cos^{-1}(3/4)$). *Hint: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$.*

Solution: Using the hint and $a + b + c = 0$,

$$0 = (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

Therefore,

$$ab + bc + ca = -\frac{1}{2}(a^2 + b^2 + c^2)$$

Note that $ab + bc + ca = \vec{u} \cdot \vec{v}$, and

$$\|\vec{u}\| \|\vec{v}\| = \sqrt{a^2 + b^2 + c^2} \sqrt{a^2 + b^2 + c^2} = a^2 + b^2 + c^2$$

Part of the point of part a) was to get you to notice this!

Since $\vec{u} \neq \vec{0}$, the denominator below is positive. We have

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|} = \frac{ab + bc + ca}{a^2 + b^2 + c^2} = -\frac{1}{2}$$

Thus, $\theta = \cos^{-1}(-1/2)$.

Problem 5 (12 pts)

Suppose $\vec{v}_1, \vec{v}_2 \in \mathbb{R}^2$ satisfy

$$\vec{v}_1 = \begin{bmatrix} 18 \\ -24 \end{bmatrix}, \quad \|\vec{v}_2\| = 6, \quad \vec{v}_1 \cdot \vec{v}_2 = 0$$

Let

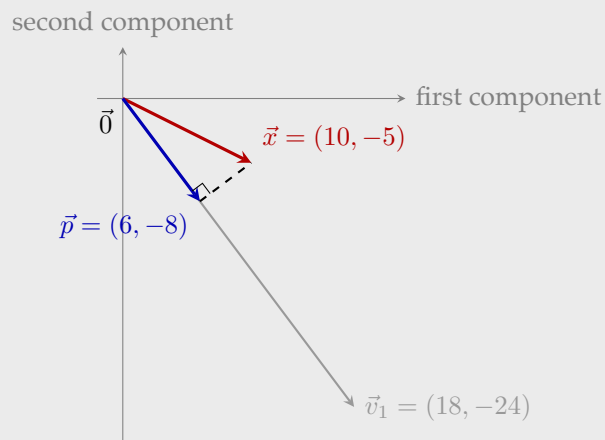
$$\vec{x} = \begin{bmatrix} 10 \\ -5 \end{bmatrix}$$

- a) (5 pts) Find $\vec{p}$, the projection of $\vec{x}$ onto $\vec{v}_1$. Give your answer as a vector with two components and no variables.

Solution: Since $\vec{v}_1 = 6 \begin{bmatrix} 3 \\ -4 \end{bmatrix}$, projecting onto $\vec{v}_1$ is equivalent to projecting onto $\begin{bmatrix} 3 \\ -4 \end{bmatrix}$.
So,

$$\vec{p} = \frac{(10)(3) + (-5)(-4)}{3^2 + (-4)^2} \begin{bmatrix} 3 \\ -4 \end{bmatrix} = \frac{50}{25} \begin{bmatrix} 3 \\ -4 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \end{bmatrix}$$

Equivalently, using $\vec{v}_1$ directly gives the coefficient $300/900 = 1/3$.



- b) (7 pts) Suppose that the first component of $\vec{v}_2$ is positive. Write $\vec{x}$ as a linear combination of $\vec{v}_1$ and $\vec{v}_2$ by filling in the two boxes below.

$$\vec{x} = \boxed{\frac{1}{3}} \vec{v}_1 + \boxed{\frac{5}{6}} \vec{v}_2$$

Solution: A vector perpendicular to $\begin{bmatrix} 3 \\ -4 \end{bmatrix}$ with positive first component has direction $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$. This direction vector has length 5, so

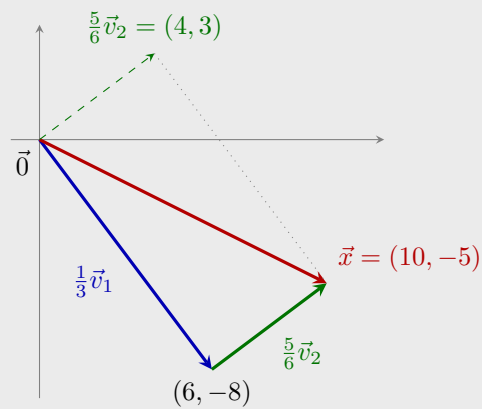
$$\vec{v}_2 = \frac{6}{5} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

From part a), $\vec{p} = \frac{1}{3}\vec{v}_1$. The remaining component is

$$\vec{x} - \vec{p} = \begin{bmatrix} 10 \\ -5 \end{bmatrix} - \begin{bmatrix} 6 \\ -8 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \frac{5}{6}\vec{v}_2$$

Therefore,

$$\vec{x} = \frac{1}{3}\vec{v}_1 + \frac{5}{6}\vec{v}_2$$



Problem 6 (19 pts)

Throughout this problem, let

$$\vec{x}_1 = \begin{bmatrix} 12 \\ -2 \\ 6 \end{bmatrix}, \quad \vec{x}_2 = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix}, \quad P = \text{span}(\{\vec{x}_1, \vec{x}_2\})$$

- a) (5 pts) Find an equation for the plane P of the form $Ax + By + Cz + d = 0$. Simplify your answer so that $A = 1$.

Solution: Because P is a span, it contains the origin, so $d = 0$. With $A = 1$, the equation is $x + By + Cz = 0$. Substituting the two spanning vectors gives

$$12 - 2B + 6C = 0, \quad 3 + 3C = 0$$

The second equation gives $C = -1$, and then the first gives $B = 3$. Thus,

$$x + 3y - z = 0$$

Both vectors lie in this plane and are linearly independent, so they span the entire plane.

For part **b)**, suppose that $\vec{x}_3, \vec{x}_4 \in \mathbb{R}^3$ and that $\text{span}(\{\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4\})$ is a **plane** in $\mathbb{R}^3$.

- b)** (5 pts) For each statement below, determine whether it is impossible, possible (but **not** guaranteed), or guaranteed to be true, given the above assumptions. The first statement has been done for you as an example.

	statement	impossible?	possible?	guaranteed?
(i)	$\ \vec{x}_3\ = 5$.	☐	☒	☐
(ii)	$\vec{x}_3$ and $\vec{x}_4$ are scalar multiples of one another.	☐	☒	☐
(iii)	$\vec{x}_4 = \vec{0}$.	☐	☒	☐
(iv)	The vectors $\vec{x}_1, \vec{x}_2, \vec{x}_3$ are linearly independent.	☒	☐	☐
(v)	$\text{span}(\{\vec{x}_3, \vec{x}_4\}) = P$.	☐	☒	☐
(vi)	$\{\vec{x}_1, \vec{x}_2\}$ is a basis for P .	☐	☐	☒

Solution: The span of all four vectors contains P and is a plane, so it must equal P . Thus, $\vec{x}_3$ and $\vec{x}_4$ both lie in P .

- (i) **Possible.** A vector in P can have length 5, but it doesn't need to.
- (ii) **Possible.** For example, choose $\vec{x}_3 = \vec{x}_1$ and $\vec{x}_4 = 2\vec{x}_1$. They don't need to be scalar multiples: we could instead choose $\vec{x}_3 = \vec{x}_1$ and $\vec{x}_4 = \vec{x}_2$.
- (iii) **Possible.** We may choose $\vec{x}_4 = \vec{0}$ because the first two vectors already span P .
- (iv) **Impossible.** Three vectors in a two-dimensional space cannot be linearly independent.
- (v) **Possible.** Choosing $\vec{x}_3 = \vec{x}_1$ and $\vec{x}_4 = \vec{x}_2$ works. Choosing both to be zero shows that this is not guaranteed.
- (vi) **Guaranteed.** By definition, $\vec{x}_1$ and $\vec{x}_2$ span P , and they are not scalar multiples of one another, so they form a basis.

Recall,

$$\vec{x}_1 = \begin{bmatrix} 12 \\ -2 \\ 6 \end{bmatrix}, \quad \vec{x}_2 = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix}, \quad P = \text{span}(\{\vec{x}_1, \vec{x}_2\})$$

Additionally, suppose that $\vec{x}_3, \vec{x}_4 \in \mathbb{R}^3$ and that $\text{span}(\{\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4\})$ is a **plane** in $\mathbb{R}^3$.

- c) (4 pts) Give one nonzero vector orthogonal to $3\vec{x}_1 - 7\vec{x}_2 + 12\vec{x}_3 - \vec{x}_4$. Your answer should have no variables. Briefly justify it.

Solution: From part a), $\vec{n} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$ is normal to P . All four vectors lie in P , so $\vec{n} \cdot \vec{x}_i = 0$ for $i = 1, 2, 3, 4$. By distributivity,

$$\vec{n} \cdot (3\vec{x}_1 - 7\vec{x}_2 + 12\vec{x}_3 - \vec{x}_4) = 0$$

Thus, one valid answer is $\begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$. Any nonzero scalar multiple of this vector also works.

For part **d)**, suppose that $\vec{x}_3, \vec{x}_4 \in \mathbb{R}^3$ but that $\text{span}(\{\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4\}) = \mathbb{R}^3$.

d) (5 pts) Which statements are **impossible**? Select all that apply.

- ☒ $\vec{x}_3$ and $\vec{x}_4$ are on P .
- ☐ The vectors $\vec{x}_3, \vec{x}_4$ are linearly independent.
- ☒ The vectors $\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4$ are linearly independent.
- ☒ There is a nonzero vector orthogonal to all four vectors.
- ☐ Every subset containing exactly three of the four vectors is a basis for $\mathbb{R}^3$.

Solution: Select **options 1, 3, and 4.**

1. **Impossible.** If both additional vectors were on P , all four vectors would span only P , not $\mathbb{R}^3$.
2. **Possible.** Choose $\vec{x}_3$ outside P and $\vec{x}_4 = \vec{x}_1$. These two vectors are independent, and all four span $\mathbb{R}^3$.
3. **Impossible.** Four vectors in $\mathbb{R}^3$ are always linearly dependent.
4. **Impossible.** A vector orthogonal to all four is orthogonal to their entire span, $\mathbb{R}^3$. In particular, it is orthogonal to itself and must be zero.
5. **Possible.** Choose $\vec{x}_3$ outside P and $\vec{x}_4 = \vec{x}_1 + \vec{x}_2 + \vec{x}_3$. The first three vectors form a basis. Replacing any one with their sum still gives a basis, since the omitted vector can be recovered by subtracting the other two from the sum.

Problem 7 (5 pts)

Consider the line L in $\mathbb{R}^3$ defined below.

$$L = \left\{ \begin{bmatrix} 6 \\ 2 \\ -4 \end{bmatrix} + t \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix} : t \in \mathbb{R} \right\}$$

a) (2 pts) True or False: L is a subspace of $\mathbb{R}^3$.

☒ True ☐ False

Solution: True. The initial vector is twice the direction vector, so

$$\begin{bmatrix} 6 \\ 2 \\ -4 \end{bmatrix} + t \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix} = (t+2) \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$$

As t ranges over $\mathbb{R}$, so does $t+2$. Hence L is the span of $\begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$, so it is a subspace of $\mathbb{R}^3$.

b) (3 pts) Consider the following two planes:

$$x + 2y + z = 0$$

$$4x + 4y + z = 0$$

Describe the set of points that lie on L and on **both** planes.

☐ There are no points in the intersection.

☒ A single point.

☐ An entire line of points.

☐ An entire plane of points.

Solution: A single point. A point on L has coordinates $x = 6 + 3t$, $y = 2 + t$, and $z = -4 - 2t$. Substituting into the two plane equations gives

$$6 + 3t = 0, \quad 28 + 14t = 0$$

Both hold exactly when $t = -2$, which gives the origin. Thus, the common intersection is $\{\vec{0}\}$.

Problem 8 (11 pts)

Consider the vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d \in \mathbb{R}^n$, where $d \geq 2$. For some scalar $t \in \mathbb{R}$, define:

$$\vec{s} = t(\vec{v}_1 + \dots + \vec{v}_d)$$

Then, for each $i = 1, 2, \dots, d$, define

$$\vec{u}_i = \vec{v}_i - \vec{s}$$

- a) (5 pts) For this part only, suppose $d = 3$. Find an expression for $\vec{u}_1 + \vec{u}_2 + \vec{u}_3$ in terms of t , $\vec{v}_1$, $\vec{v}_2$, and $\vec{v}_3$. Note that your answer **cannot** involve $\vec{s}$.

Solution: For $d = 3$, $\vec{s} = t(\vec{v}_1 + \vec{v}_2 + \vec{v}_3)$. Therefore,

$$\begin{aligned}\vec{u}_1 + \vec{u}_2 + \vec{u}_3 &= (\vec{v}_1 - \vec{s}) + (\vec{v}_2 - \vec{s}) + (\vec{v}_3 - \vec{s}) \\ &= \vec{v}_1 + \vec{v}_2 + \vec{v}_3 - 3t(\vec{v}_1 + \vec{v}_2 + \vec{v}_3) \\ &= \boxed{(1 - 3t)(\vec{v}_1 + \vec{v}_2 + \vec{v}_3)}\end{aligned}$$

- b) (4 pts) Find a value of t for which $\vec{u}_1, \dots, \vec{u}_d$ are **guaranteed** to be linearly dependent, regardless of the original vectors. Give your answer as an expression in terms of n , d , and/or constants.

Solution: As in part a), observe what happens when we sum all of the $\vec{u}_i$. This is a linear combination of the $\vec{u}_i$, with every coefficient equal to 1:

$$\begin{aligned}\vec{u}_1 + \dots + \vec{u}_d &= (\vec{v}_1 - \vec{s}) + \dots + (\vec{v}_d - \vec{s}) \\ &= \vec{v}_1 + \dots + \vec{v}_d - dt(\vec{v}_1 + \dots + \vec{v}_d) \\ &= (1 - dt)(\vec{v}_1 + \dots + \vec{v}_d)\end{aligned}$$

To make this equal to $\vec{0}$ regardless of the original vectors, set

$$1 - dt = 0 \implies \boxed{t = \frac{1}{d}}$$

For this value of t , the sum is $\vec{0}$. Since the coefficients in this linear combination are all 1, they are not all zero. By the definition of linear independence, the $\vec{u}_i$ are therefore linearly dependent.

c) (2 pts) Suppose $t > 0$. Fill in the $\boxed{??}$ with the relationship that is **guaranteed**:

$$t(\|\vec{v}_1\| + \|\vec{v}_2\| + \cdots + \|\vec{v}_d\|) \quad \boxed{??} \quad \|\vec{s}\|$$

☐ $>$ ☒ $\geq$ ☐ $=$ ☐ $<$ ☐ $\leq$

Solution: Since $t > 0$, the triangle inequality gives

$$t(\|\vec{v}_1\| + \cdots + \|\vec{v}_d\|) \geq t\|\vec{v}_1 + \cdots + \vec{v}_d\| = \|\vec{s}\|$$

The guaranteed relationship is $\boxed{\geq}$. Equality can occur, for example, when all the original vectors are the same nonzero vector, so a strict inequality is not guaranteed.